Jaydip Singh, Ph.D.

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LinkedIn | GitHub | Portfolio | Research Profile

Professional Summary

Ph.D.-qualified Machine Learning Scientist and Lead GenAI Engineer with 10+ years of combined academic and industry experience spanning scientific computing, software/hardware engineering, and end-to-end AI/ML systems. Specializes in designing and deploying production-grade agentic AI on AWS — using Amazon Bedrock, Bedrock AgentCore (agent runtimes, memory, multi-agent orchestration), and the Strands Agents framework — to scale enterprise solutions across business domains. Deep expertise in multi-agent architectures, multi-LLM/SLM model selection, multi-index semantic retrieval and context engineering, RAG, knowledge graphs, and semantic/metadata layers (data catalog, lineage, ontology), with strong command of ML tradeoffs between deterministic and reasoning-driven logic, benchmarking/evaluation, and MLOps/observability. Lead Engineer (GenAI) at Apexon, delivering client-facing, enterprise-scale solutions on AWS. Also an independent researcher on efficient small and multimodal language models.

Independent research: two 2026 preprints on efficient small and multimodal language models trained on constrained hardware — a 23M physics SLM that outperforms GPT-2 Large (774M) by 69%, and a multimodal medical-physics visual-QA model reaching 89% on a simulation-generated benchmark. Portfolio: github.com/JaydipSingh/portfolio.

Technical Skills

- **AI / GenAI / Agentic:** Amazon Bedrock, Bedrock AgentCore (agent runtimes, memory, multi-agent orchestration), Strands Agents framework, Azure OpenAI, LangChain, LangGraph, MCP, RAG, multi-agent systems, multi-LLM/SLM model selection, context engineering, multi-index semantic retrieval, prompt engineering, embeddings/vector retrieval, knowledge graphs, ontology/semantic layer, synthetic data generation, LLM benchmarking & evaluation.
- **ML / Analytics:** Machine Learning, Deep Learning, SageMaker, forecasting (LSTM, P10/P50/P90 probabilistic), feature engineering, CNNs, Monte Carlo simulation, MLOps / model observability, statistical modeling.
- **Programming:** Python (NumPy, Pandas, PyTorch, TensorFlow, Scikit-learn), C/C++, SQL, Bash/Shell.
- **Cloud / Platform:** AWS Lambda, API Gateway, DynamoDB, S3, Cognito, Secrets Manager, WAF, CloudWatch, EventBridge, Step Functions, SNS, ECS/Fargate, ECR, CodeBuild, SSM, KMS, Redshift, SageMaker, Glue ETL, Bedrock, Bedrock AgentCore; Azure Functions, Azure AI Search.
- **DevOps / IaC:** AWS CDK, Terraform, Docker (ARM64), GitHub Actions, CI/CD, OIDC, IAM (least-privilege), observability (CloudWatch).
- **Frontend / Backend / Data:** React, TypeScript, FastAPI, Flask, REST/WebSocket APIs, Node.js; PostgreSQL, Chroma, Pinecone, Neo4j AuraDB, Redshift, S3 Data Lake.
- **Simulation/Design:** GEANT4 (detector development), COMSOL (chip/semiconductor simulation), SolidWorks (hardware design), LaTeX.

Selected Projects

- **SLM Physics Assistant — Domain-Specialized Small Language Model** (Rust, Python, PyTorch, MPS)
  - Built an end-to-end, reproducible stack — Rust BPE tokenizer, custom decoder-only transformer (RoPE, RMSNorm, SwiGLU), LoRA fine-tuning, and multi-metric evaluation — trained entirely on a single Apple M3 Pro laptop.
  - A 23M-parameter physics-specialized model scores 0.55 on a physics QA benchmark, outperforming GPT-2 Large (774M) by 69% despite being 33× smaller.
  - Diagnosed and fixed an early end-of-sequence collapse failure mode via EOS-free document packing; added automated tests and CI. Research paper (preprint; arXiv forthcoming).
  - GitHub Repo
- **PhysVision-SLM — Multimodal Medical-Physics Visual QA** (PyTorch, SigLIP, Qwen2, Transformers, MPS)
  - Trained a multimodal SLM entirely on simulation-generated data (an analytical image simulator provides exact ground truth, so no human annotation is required).
  - A frozen SigLIP + Qwen2-0.5B configuration reaches 89.0% on a 35,596-pair visual-QA benchmark (vs. 25% random); a controlled lite-vs-full comparison attributes the gap to the vision encoder.
  - Built the dataset generator, two-stage training, and log-probability-ranking evaluation with diagnostic checks. Research paper (preprint; arXiv forthcoming).
  - GitHub Repo
- **Agentic Video Generator** (FastAPI, LangGraph, Redis, Postgres, ffmpeg, Streamlit, React)

- Built an end-to-end multi-agent video generation pipeline that transforms story prompts into rendered MP4 outputs via a checkpointed LangGraph workflow.
- Implemented asynchronous API + worker architecture with Redis queue and Postgres-backed job state for reliable long-running generation tasks.
- Added node-level checkpoint persistence (Redis cache + Postgres durable store) with resume endpoint support to recover from mid-pipeline failures.
- Delivered two production-ready UIs (Streamlit and React) with submission forms, status polling, resume action, and in-app video preview.
- GitHub Repo

Professional Experience

Lead Engineer (GenAI) – Apexon (Client: AWS), Mumbai, India Dec 2025 – Present

- **Congress Operations Management Platform** (07/2026 – Present): Led end-to-end design and delivery of a cloud-based operations platform unifying attendee processing, booth staffing, report matching, and operational analytics. Built Bedrock-based agents (Supervisor on Bedrock AgentCore + Strands Agents framework), a React/TypeScript frontend and Python serverless backend, with Terraform/AWS CDK infrastructure, Cognito/IAM role-based security, and Docker/ECR + GitHub Actions deployment pipelines (ARM64).
- **AI-Powered Clinical Trial Supply Chain Predictive Analytics Platform** (04/2026 – Present): Architected a 7-layer AWS serverless platform predicting drug demand and detecting stockout/expiry risk for a pharmaceutical client. Built SageMaker LSTM demand forecasting (P10/P50/P90, MAPE <20%), 5 specialist Amazon Bedrock agents, event-driven ingestion (EventBridge/Lambda/Glue), and automated compliance (TOR) reporting via Step Functions + Bedrock Claude — cutting reporting from days to under 60 seconds, saving 20+ hours/week and preventing $400K+/year in losses per trial program.
- Applied ML judgment on deterministic vs. reasoning-driven logic and multi-LLM/SLM model selection, with systematic benchmarking of agent accuracy, latency, and cost against enterprise SLAs; operationalized MLOps/observability (drift detection, retraining triggers, CloudWatch metrics).

Gen AI Technical Lead (IC) – FutureSoft (Client: PwC BSC), Mumbai, India Jul 2025 - Nov 2025

- Leading Generative AI projects for PwC, including enterprise-scale chatbot/agent development, RAG-based automation, and client-facing AI solutions.
- Architecting and deploying AI/ML pipelines with focus on LLMs, LangChain, and cloud-based MLOps practices.
- Mentoring cross-functional teams on GenAI adoption, responsible AI practices, and production-grade deployments.
- Collaborating with stakeholders to translate business requirements into scalable AI-driven solutions.

Postdoctoral Research Scientist (Data Scientist) – UC Davis, USA Jan 2024 – Dec 2024

- Developed AI agents for real-time data monitoring and anomaly detection using RAG, BERT, Flan-T5, and GPT.
- Integrated Pinecone vector DB for log ingestion, enabling advanced reasoning and context retention.
- Applied ML algorithms (PCA, DBSCAN, CNNs) for image classification and 3D reconstruction.
- Built Monte Carlo–based models for optimizing detector performance.

Senior Software Engineer – Nextgen Clearing Pvt Ltd., Mumbai, India Nov 2021 – May 2023

- Engineered telecom billing systems using Python, C++, and PL/SQL.
- Refactored legacy modules, boosting performance and maintainability.
- Coordinated development and debugging with cross-functional teams.

AI Research Collaborator – University of Lucknow & Fermilab, USA Mar 2019 – Oct 2021

- Developed CNN/GenAI models for particle physics image reconstruction (LArTPC).
- Optimized classification and detector performance using deep learning.
- Performed large-scale data analysis using PyTorch and Scikit-learn.

Research Fellow (Scientific Software Developer) – India-based Neutrino Observatory, Lucknow University Aug 2014 – Mar 2019

- Created GEANT4-based simulation software for experimental physics.
- Integrated ML models to enhance detector signal recognition and efficiency.
- Deployed on Fermi Cloud (AWS/GCP-like) infrastructure for distributed computing.

Education

- **Ph.D. in Particle Physics**, University of Lucknow, India 2019 – 2022
- **M.Sc. in Physics**, Indian Institute of Technology Bombay, India 2011 – 2013
- **B.Sc. in Physics, Mathematics & Computer Science**, Ewing Christian College, Allahabad 2007 – 2010

Research Publications

Extensive publication record in high-energy physics and AI/ML research.
Full list available at: inspirehep.net/authors/1591917

Preprints / Under Review (Independent Research)

- **A Reproducible Systems Stack for Domain-Specialized Small Language Models: Tokenizer, Training, and Evaluation on Constrained Hardware** (2026). Preprint; arXiv forthcoming. Code: github.com/JaydipSingh/slm-physics-assistant.
- **Simulation-Generated Data for Multimodal Small Language Models: A Constrained-Hardware Recipe for Medical Physics Visual Question Answering** (2026). Preprint; arXiv forthcoming. Code: github.com/JaydipSingh/physvis-slm.

Selected AI/ML-Enhanced Publications (Key Contributions)

- **Separation of track- and shower-like energy deposits in ProtoDUNE-SP using a convolutional neural network**, Eur. Phys. J. C 82 (2022).
  Technologies: Convolutional Neural Network (CNN), supervised classification, image-based detector data.
  Contributed to integration of CNN model into the reconstruction workflow and optimization with real and simulated data.
  inspirehep.net/2060793
- **Highly-parallelized simulation of a pixelated LArTPC on a GPU**, JINST 18 (2023).
  Technologies: GPU parallelization (CUDA), high-performance computing, simulation pipelines.
  Developed and optimized GPU-accelerated simulation kernels and integrated with experiment frameworks.
  inspirehep.net/2620145
- **Neutrino interaction vertex reconstruction in DUNE with integrated U-ResNet neural network** (2025).
  Technologies: U-ResNet (deep residual U-Net), hit-level classification, hybrid ML pipeline integration.
  Integrated U-ResNet into vertex reconstruction pipeline, ensuring efficient interfacing and runtime optimization.
  inspirehep.net/2878306

Certifications & Test Scores

- AI Engineer Production Track: Deploy LLMs & Agents at Scale (Udemy)
- AI Engineer Agentic Track: The Complete Agent & MCP Course (Udemy)
- Generative AI with Large Language Models (AWS, DeepLearning.AI)
- IIT-JAM (Physics): All India Rank 263 (2011)
- NGPE (IAPT): Top 10 percentile (2010)

Awards & Fellowships

- Rajendra Raja Fellowship – Fermilab, USA
- INO Fellowship – DST, India

Extracurricular Activities

- Winner: 4x100m Relay (IIT-Bombay & ECC), Basketball/Volleyball tournaments
- Finalist: 400m Sprint, Annual college meets (2008–2010)